Haptic-Guided Shared Autonomy for Full-Scale Vehicle Teleoperation

Design and integration of force-feedback steering and motion cueing for remote vehicle operation

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Motivation

Remote vehicle teleoperation is essential in hazardous and inaccessible environments [3], yet current interfaces provide **no vestibular or haptic feedback** [5], allocate control authority in a binary fashion [6], and consequently **elevate cognitive workload** [1]. This work integrates force-feedback steering, motion cueing, and continuous authority blending to address these gaps.

Objectives

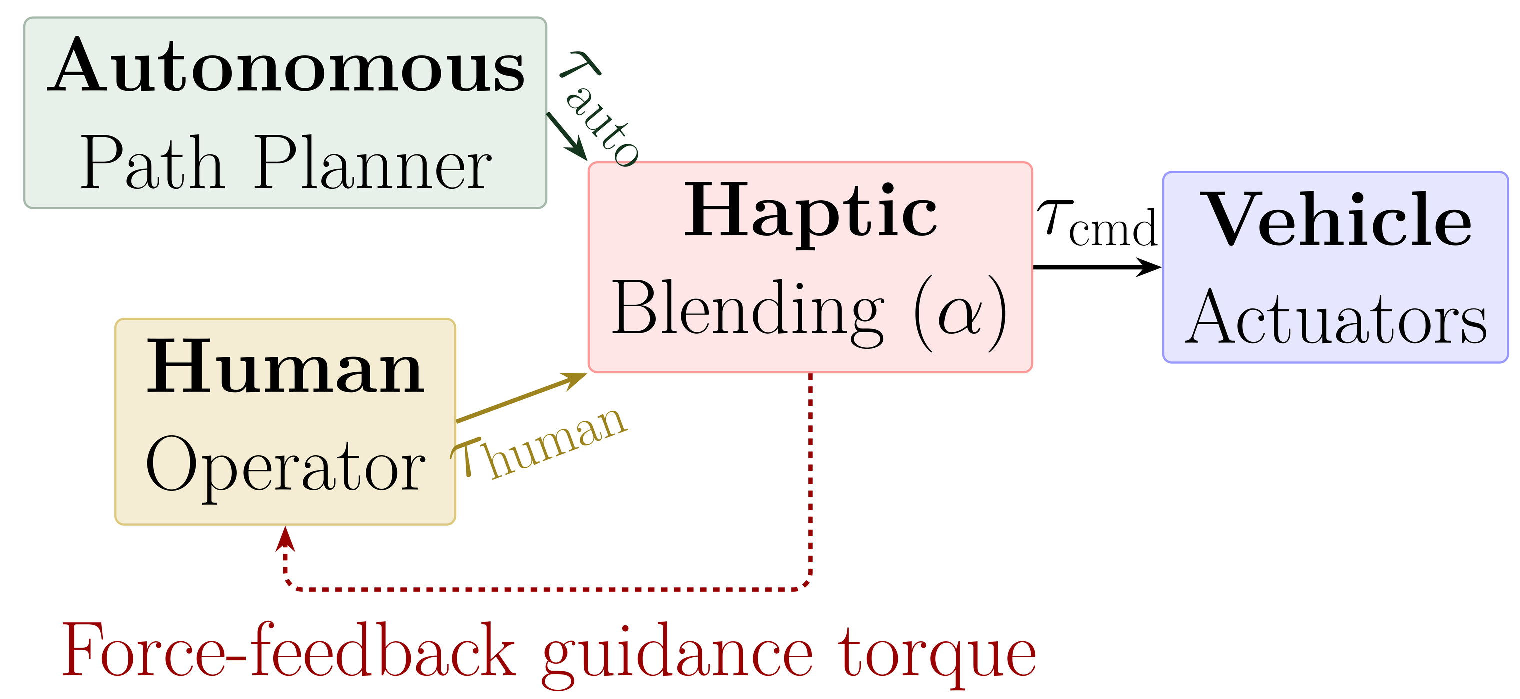
This work addresses these limitations through three integrated subsystems:

1. A **haptic shared-control interface** that blends operator and autonomous steering commands through continuous force-feedback torques [1, 2]
2. An **IMU-driven motion cueing system** that reproduces vehicle dynamics on a multi-actuator platform to restore vestibular feedback [5]
3. A **full-scale drive-by-wire vehicle platform** with distributed CAN bus control and hardware watchdog safety [3]

Vehicle Platform

Parameter	Specification
Chassis	Electric UTV, 159 kg, 1.23 m wheelbase
Powertrain	48 V lithium, 1.8 kW brushless motor
Actuation	Drive-by-wire steer, throttle, brake
Bus	CAN 2.0B, 250 kbps, 4× Teensy 4.1
Compute	NVIDIA Jetson Orin AGX
GPS/IMU	Xsens MTi-680G, RTK ±5–10 cm, 100 Hz
LiDAR	Velodyne VLP-16, 16-channel
Camera	Intel RealSense D435 (RGB-D)

Haptic Shared Control — Method



A Pure Pursuit controller [4] generates a reference steering angle from a planned path. Following the haptic shared-control paradigm [1], the deviation between the reference and current steering angle produces a guidance torque through a virtual spring model:

$$\tau_{\text{auto}} = K_s \cdot \delta_{\text{error}}$$

The operator perceives this torque as force feedback on the steering wheel [2]. The commanded actuator torque is computed as a weighted sum of the autonomous and operator inputs (input-mixing architecture) [6]:

Torque Blending Law

$$\tau_{\text{cmd}} = \alpha\,\tau_{\text{auto}} + (1-\alpha)\,\tau_{\text{human}}, \quad \alpha \in [0,1]$$

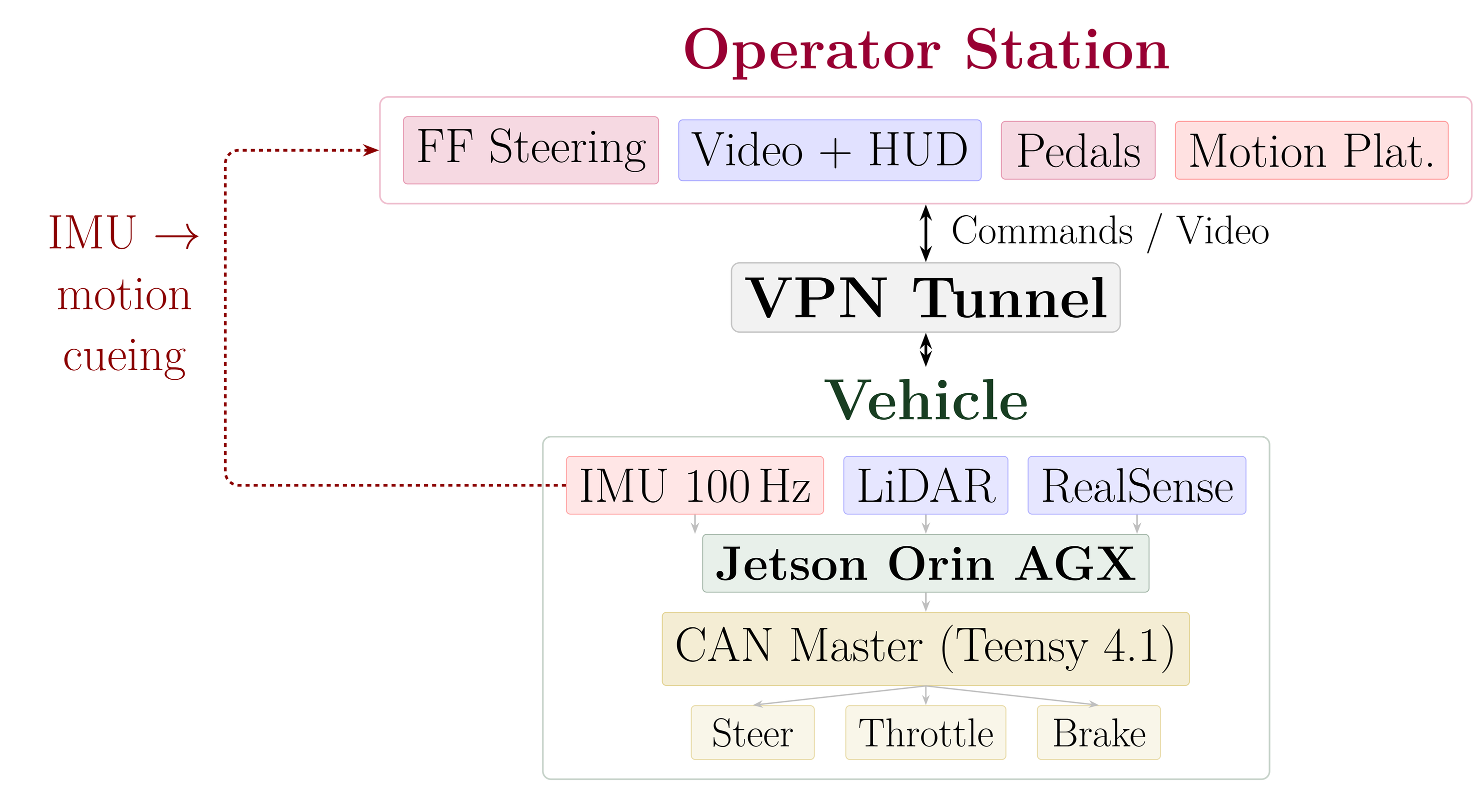
- $\alpha = 0$: full manual control $\longleftrightarrow$ $\alpha = 1$: full autonomy
- α is **operator-adjustable**; at any $\alpha < 1$ the operator retains the ability to override the autonomous input [6]
- Adaptive lookahead distance: $l = \text{clip}(K_{dd} \cdot v,\ l_{\min},\ l_{\max})$ [4]

System Characterization

Parameter	Design Target	Status
Control round-trip latency	< 50 ms	Under characterization
IMU-to-actuator latency	< 100 ms	Under characterization
Haptic rendering rate	≥ 1 kHz	Verified
CAN bus command rate	100 Hz	Verified
Watchdog timeout	500 ms	Configured
Localization accuracy (RTK)	±5–10 cm	Verified

Items marked *Verified* have been confirmed through bench-level measurement.

System Architecture



Vehicle IMU data streams over VPN to the operator station, where four suspension actuators reproduce **pitch**, **roll**, and road-surface vibration. Distributed CAN nodes close actuator control loops locally at 100 Hz.

Planned Evaluation

A within-subjects experiment ($n \geq 10$) will compare **direct teleoperation** against **haptic shared autonomy** using cross-track error, task completion time, NASA-TLX workload [3], and operator override frequency. Bench characterization of haptic and motion-cueing latency will precede outdoor driving trials.

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References

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